QP CODE: 24900110

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION
NOVEMBER 2024
First Semester Multi-Disciplinary Course - MG1MDCFTQ100 - FOOD SPOILAGE
(2024 ADMISSION ONWARDS)

Duration: 2 Hours Max Marks 35

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)
Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.
[Learning Domain][CO No(s)]

Part A Short Answer Type Questions Answer any 5 questions
Each questions carries 1 mark

1 Define water activity [U] [1]
Water activity (often abbreviated as *aw*) is a measure of the availability of water in a substance for microbial growth, chemical reactions, and physical processes like drying or crystallization. It quantifies the relative humidity of the air in equilibrium with a material, expressed as a ratio of the vapor pressure of the water in the material to the vapor pressure of pure water at the same temperature
2 How do yeasts cause spoilage in food products? [U] [2]
Yeasts, particularly *Saccharomyces cerevisiae* and other species, are capable of fermenting sugars in food, converting them into alcohol and carbon dioxide. This process can lead to:alcohol production and carbon dioxide production
3 What are the main types of spoilage? [U] [3]
Microbial spoilage, physical spoilage, chemical spoilage and physical chemical spoilage
4 Give examples of pests, insects, and rodents contributing to spoilage. [U] [4]
Rodents (e.g., Mice, Rats)
Insects such as *grain beetles* (e.g., *Tribolium* species), *mealworms* (larvae of darkling beetles), *Indian meal moths*, and *rice weevils* consume food and reproduce in it.
Beetles (e.g., *Flour Beetles*, *Cigarette Beetles*)
5 What are the visible signs of 'thamnidium taint'? [U] [5]
Fuzzy or Powdery Mold Growth,Color Changes and Sensory changes
6 Why is light exposure a concern for food storage? [U] [1]
Light, particularly ultraviolet (UV) light, can degrade or break down several important nutrients in food, leading to nutrient loss
Key nutrients that are particularly sensitive to light include:

- Vitamins,Carotenoids, Folate

The degradation of these nutrients can lead to a decrease in the overall nutritional value of the food, which can affect health benefits and reduce the food's appeal.

7 Define O-R Potential of food [U] (3)

In the context of food, O-R potential refers to the balance between oxidation (electron loss) andreduction (electron gain) processes occurring in the food's environment. It is influenced by factors such as: Presence of oxygen,Microbial activity and Food composition**(5 × 1 = 5)**

Part B Short Essay Type Questions Answer any 4 questions

Each question carries 5 marks

8. Discuss the role of pH in food spoilage. How does it affect microbial growth, enzymatic activity, and overall food safety?[U] [1]

The pH value of a food is a direct function of the free hydrogen ions present in that food. Acids present in foods release these hydrogen ions, which give acid foods their distinct sour flavor. Thus, pH may bedefined as a measure of free acidity. Microorganisms, including yeasts, molds, and bacteria, are sensitive to a food's pH. Very low or very high pH values will prevent microbial growth. Extreme pH affects the structure of all macromolecules. The hydrogen bonds holding together strands of DNA break up at high pH. Lipids are hydrolyzed by an extremely basic pH The proton motive force responsible for production of ATP in cellular respiration depends on the concentration gradient of H^+ across the plasma membrane (see Cellular Respiration). If H^+ ions are neutralized by hydroxide ions, the concentration gradient collapses and impairs energy production. But the component most sensitive to pH in the cell is its workhorse, the protein. Moderate changes in pH modify the ionization of amino-acid functional groups and disrupt hydrogen bonding, which, in turn, promotes changes in the folding of the molecule promoting denaturation and destroying activity.

9 What are mycotoxins, and how are they related to mold spoilage? [U] [2]

Mycotoxins are toxic compounds produced by certain molds found in food and feed, posing health risks to humans and animals. These toxins can contaminate crops both pre- and post-harvest, leading to exposure through consumption of contaminated products. Mycotoxins exert their toxic effects through complex biochemical interactions within the body. These compounds can disrupt cellular processes by interfering with DNA replication and protein synthesis, leading to cellular damage and death. For example, aflatoxins form reactive intermediates that bind to DNA, resulting in mutations and an increased risk of cancer, particularly liver cancer. This interference with genetic material is a hallmark of many mycotoxins and underscores their potential to cause long-term health issues.

10 List the sources of contamination of meat. [U] [3]

Microbial contamination mainly occurs on the surface of meat, which can lead to spoilage but also to the transmission of foodborne pathogens . This is a significant concern for the meat industry and consumers. The source of meat contamination varies, with the main source being the animals themselves. During slaughter, the carcasses can become contaminated by the hides of cattle, the fleece of sheep, the skin of pigs, or the feathers of poultry that can all carry animal feces, soiling, and animal feed on them, while the spillage of the intestinal material of the slaughtered animals during the evisceration procedure can also lead to contamination of the carcasses . Other sources that are responsible for cross-contamination of the carcasses with microorganisms are the processing environment, the equipment used and the workers, while insects, rodents and birds can also carry and

transmit microorganisms to meat. Furthermore, microbial contamination and growth may also take place after processing, and during storage and distribution of meat products .

11 How does enzymatic spoilage differ from microbial spoilage? [An] [4]

Enzymatic spoilage is caused by the natural enzymes present within the food itself. These enzymes are proteins that catalyze biochemical reactions and can cause changes in the food's texture, flavor, and nutritional value over time. WhereasMicrobial spoilage is caused by microorganisms such as bacteria, molds, and yeasts that grow and multiply on the food. These microorganisms thrive on the nutrients in the food, breaking them down and often producing metabolic byproducts that cause spoilage.

12 Discuss about shellfish spoilage. [E] [5]

Shellfish spoilage is a specific type of food deterioration that can be caused by several factors, including enzymatic activity, microbial contamination, and environmental conditions. Shellfish are highly perishable, and their spoilage can pose serious health risks if not properly handled, stored, or cooked.

Key Causes of Shellfish Spoilage:

1. Microbial Spoilage:2. Enzymatic Spoilage:3. Physical and Chemical Spoilage:

Signs of Spoiled Shellfish:

- Off-smell, Texture Changes, DiscolorationOpen Shells and Gassiness.

13 What is water activity (aw), and how is it measured? [U] [2]

Water activity (aw) is a measure of the available water in a substance that is free to interact with other components (such as solutes, microbes, or enzymes) and participate in chemical and biological reactions. In simple terms, water activity indicates how much of the water in a food, material, or substance is not bound to other molecules (like salts, sugars, or proteins) and is available for microbial growth, chemical reactions, and enzymatic activity.For example, lower water activity (such as in dried foods) helps inhibit the growth of many microorganisms (including bacteria, mold, and yeast) and slows down spoilage and degradation processes.

Water Activity Scale:

- Water activity is measured on a scale from 0 to 1, where:
    - 0 means no available water (completely dry),
    - 1.0 means pure water (with 100% of the water available

14. Describe microbial spoilage and its primary causes. [An] [4]

Microbial spoilage refers to the deterioration of food or other materials due to the growth and activity of microorganisms, including bacteria, yeasts, and molds. These microorganisms break down the food's components (e.g., proteins, carbohydrates, fats), leading to undesirable changes in texture, color, flavor, odor, and sometimes nutritional value. Microbial spoilage can result in the food becoming unsafe to consume, especially if pathogenic organisms are involved.

Types of bacteria involved in spoilage:

- Psychrotrophic bacteria, Mesophilic bacteria.Thermophilic bacteriaLactic acid fermentation
- LipolysisGas production

2. Yeastsand3.Mould

Factors That Promote Microbial Spoilage:

Microbial spoilage is influenced by several environmental and intrinsic factors, which provide favorable conditions for microbial growth and activity.

- TemperatureMoisture, pH and Nutrients

(4 × 5 = 20)

Part C Long Essay Type Questions Answer any 1 question
Each question carries 10 marks

15 Discuss about the role of nutrients, water activity and inhibitory substances in microbial growth [An] [1]

Microbial growth is a complex process influenced by various environmental factors. Among the most important factors that control the growth of microorganisms are nutrients, water activity (aw), and inhibitory substances. Each of these factors plays a crucial role in determining whether microorganisms will thrive, remain dormant, or be inhibited.

1. Nutrients- Macronutrients, Micronutrients and Growth factors
2. Water Activity (aw) 3. Inhibitory Substances

16 Describe what extrinsic factors are in the context of food spoilage. Discuss how these factors differ from intrinsic factors. [An] [2]

In the context of food spoilage, extrinsic factors and intrinsic factors are two categories of environmental conditions that influence the growth of microorganisms and the spoilage process of food.

Extrinsic Factors- Temperature, Relative HumidityAtmosphere (Oxygen, Gases)Storage Conditions and Time

Intrinsic Factors

Intrinsic factors, on the other hand, are properties inherent to the food itself. These factors are related to the composition and structure of the food, which can influence its susceptibility to microbial growth and spoilage. The main intrinsic factors include:

pH (Acidity/Alkalinity),Water Activity (aw)Nutrient ContentAntimicrobial Constituents , Structure and Composition and Preservatives.

Differences Between Extrinsic and Intrinsic Factors

- Origin: Extrinsic factors come from the environment surrounding the food (temperature, humidity, packaging, etc.), whereas intrinsic factors are inherent to the food itself (pH, water activity, nutrient content).
- Control: Extrinsic factors are often more easily controlled during food storage and handling (e.g., refrigeration, proper packaging), while intrinsic factors are fixed properties of the food that cannot be easily changed (e.g., the natural acidity of citrus fruits).
- Effect on Microbial Growth: Extrinsic factors primarily influence the rate of microbial growth, whereas intrinsic factors can either directly support or inhibit the types of microorganisms that can grow on the food. For example, a high-water-activity food stored at room temperature will spoil faster than a low-water-activity food, even if both have similar nutrient content.

17 Describe the spoilage in meat and meat products [E] [3]

Spoilage in meat and meat products is a significant concern in food safety, as it not only affects the quality and nutritional value of the product but can also lead to foodborne illnesses if pathogenic

microorganisms are involved. Meat is an excellent medium for microbial growth due to its high protein and moisture content, making it highly susceptible to spoilage under improper storage conditions.

Microbial Spoilage

Bacteria, Fungi: Molds and Yeasts ,Viruses and Protozoa:

Factors Contributing to Spoilage in Meat

Spoilage in meat is influenced by a combination of intrinsic and extrinsic factors, including:

Intrinsic Factors, Spoilage Indicators in Meat

Spoilage in meat is often characterized by the following sensory changes:

Odor:Colour ,Texture: and Slime Formation:

Methods to Control Spoilage in Meat

To prevent or delay spoilage in meat and meat products, several methods can be applied:

1. Refrigeration and Freezing:
2. Curing, Salting, and Smoking:
3. Packaging:
4. Acidification (Fermentation):
5. Antimicrobial Treatments:

$(1 \times 10 = 10)$